

A Proof-of-Concept Study: Enhancing Microbial Volatile Profiles for Rapid Screening of *Pseudomonas* in Milk

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INTRODUCTION

Pseudomonas spp. are psychrotrophic spoilage bacteria that grow in refrigerated raw milk and produce heat-stable proteases such as AprX [1]. Although ultra-high-temperature (UHT) processing destroys bacterial cells, residual enzymes can remain active, causing bitterness, sedimentation, and age gelation during storage [1]. However, routine dairy testing is not specific to *Pseudomonas*, while dedicated microbiological testing can be costly and time-consuming, with no rapid, non-invasive screening approach currently available [2].

Solid-phase microextraction-gas chromatography–time-of-flight mass spectrometry (SPME-GC–TOFMS) can detect microbial volatile organic compounds (VOCs), but naturally occurring biomarkers may be too low in abundance without bacterial growth [3,4]. As VOCs arise from microbial metabolism, enhancing relevant metabolic pathways may boost their production. This *proof-of-concept* study therefore investigated metabolic boosting to enhance targeted microbial volatile biomarkers and improve SPME-GC–TOFMS-based *Pseudomonas* screening in milk.

METHODS

Four proteolytic *Pseudomonas* isolates from raw milk were spiked (i.e., total 4 CFU/mL) into sterile UHT milk and incubated at 25°C for 48 h with or without a proprietary metabolic booster; uninoculated UHT milk with and without the booster served as controls. Post-incubation, *Pseudomonas* loads were enumerated to assess the effects on growth, and all samples were stored at –20°C until subsequent SPME-GC–TOFMS analysis, with a focus on booster-responsive VOCs. The same approach was applied to raw milk from six farms, with *Pseudomonas* counts ranging from <10 to >1,000 CFU/mL. Booster details are withheld for intellectual property (IP) protection. Log₂ fold changes were calculated between boosted and non-boosted samples, with $q < 0.05$ considered significant.

RESULTS AND DISCUSSION

Post-incubation *Pseudomonas* counts were comparable with and without the metabolic booster (2.32×10^8 and 2.58×10^8 CFU/mL, respectively). In *Pseudomonas*-spiked UHT milk, metabolic boosting generated three distinct putative biomarkers, with significantly increased abundance (log₂ fold changes of 6.97, 7.48, and 8.05 for Biomarkers 1–3; $q < 0.05$), while minimal differences were observed in non-boosted samples (Figure 1). The same biomarkers were detected in raw milk, where abundance increased significantly across all

Pseudomonas levels tested ($q < 0.05$), including samples below the detection limit of traditional microbial testing (<10 CFU/mL). Metabolic boosting therefore enabled detection of microbial volatile biomarkers at very low *Pseudomonas* levels without requiring the bacteria to first grow to higher numbers for naturally occurring biomarkers to reach detectable levels, demonstrating the potential of this strategy for GC–TOFMS-based screening in complex milk samples.

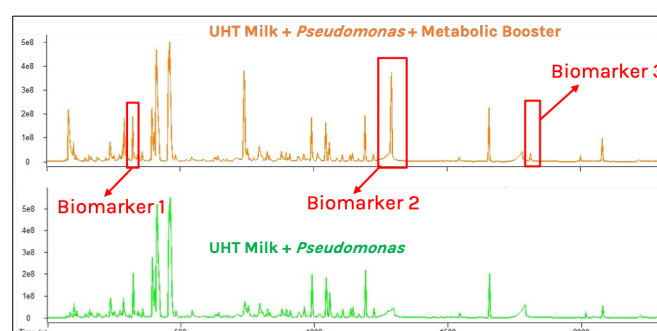


Figure 1 GC–TOFMS chromatograms showing three putative biomarkers enhanced by metabolic boosting in *Pseudomonas*-spiked UHT milk (25°C, 48 h). Biomarker identities and booster details withheld due to ongoing IP agreements with industry.

CONCLUSIONS

This study presents a novel approach that boosts microbial volatile biomarkers to improve rapid, non-invasive *Pseudomonas* screening by SPME-GC–TOFMS, with minimal sample preparation and solvent use. With further validation, it may support dairy quality control, earlier decision-making, shelf-life management, and future on-site applications. Future work is required to validate the putative biomarkers, assess their specificity to *Pseudomonas* in mixed microbiota, and evaluate sensitivity across microbial loads.

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